

Akarsh Kothakonda
Sr. Python Developer
Akarshkushal@gmail.com
+1 443-494-8278

PROFESSIONAL SUMMARY:

- Senior Full Stack Developer with 10+ years of experience delivering scalable, high-performance solutions using advanced Python and Angular frameworks. Proven expertise in enterprise-level application development on AWS and other cloud platforms. Adept at architecting end-to-end solutions, agile project management, and CI/CD automation in fast-paced, remote work environments. Passionate about integrating robust back-end logic with modern, responsive Angular interfaces while driving innovation in financial and capital market contexts.
- Strong proficiency in backend development with Python (Django, Flask, FastAPI), cloud-based deployments using AWS (EC2, RDS, S3, Lambda), and performance tuning for high-traffic applications.
- Proven ability to collaborate with clients and cross-functional teams, ensuring smooth project execution. Experienced in automating network infrastructure using Python and Ansible, with hands-on experience in SNMP, SSH, IAM policies, and API security. Basic understanding of routing protocols such as BGP, OSPF, and ISIS for configuring and troubleshooting network devices.
- Proven ability to design scalable microservices architectures and ensure secure, high-performance applications. Skilled in developing and optimizing high-performance APIs using Python (Django, Flask, FastAPI) and AWS services (EC2, Lambda, S3, RDS). Experienced in automating deployments, monitoring system performance, and implementing scalable infrastructure to enhance reliability and efficiency.
- Expertise in Python, Django, Flask, Fast API, and front-end technologies like React.js, Vue.js, and Angular.
- Skilled in cloud platforms like AWS and Azure and utilizing services like EC2, S3, RDS, Lambda, and Azure Functions to build scalable and reliable applications.
- Proficient in database management with MySQL, PostgreSQL, DynamoDB, MongoDB, and NoSQL databases, as well as implementing data processing and machine learning workflows using libraries like NumPy, Pandas, and TensorFlow.
- Strong experience building RESTful APIs, microservices, and event-driven architectures and working with tools like Docker, Kubernetes, Jenkins, and Terraform for DevOps automation.
- Proven ability to drive high-performance teams through agile development practices while ensuring applications' security, scalability, and maintainability.
- Strong communicator with a collaborative approach, driving cross-functional teamwork to develop innovative solutions and improve productivity in high-pressure environments.
- Highly adaptable and quick to learn, demonstrating a proactive problem-solving mindset while staying current with emerging technologies to optimize performance and ensure client satisfaction.

TECHNICAL SKILLS:

Programming & Backend: Python (Django, Flask, FastAPI), AsyncIO, Multithreading, Concurrency

Frontend Technologies: Angular, Typescript, Javascript, HTML5, CSS3

API Development: RESTful APIs, GraphQL, API Gateway

Cloud & Infrastructure: AWS (Redshift, Glue, EMR, EC2, RDS, Lambda, API Gateway, CloudFormation, S3), Kubernetes, OpenShift

SQL & Database Management: PL/SQL, MySQL, PostgreSQL, Redshift, SQL Server, DynamoDB, MongoDB

Data Processing: PySpark (ETL, Data Transformation, Distributed Computing), SQL (PostgreSQL, MySQL, Redshift, Snowflake), Apache Airflow, Pandas, NumPy

DevOps & CI/CD: Jenkins, Ansible, AWS CodePipeline, Terraform, GitHub Actions, Quartz Scheduler, Artifactory

Security & Authentication: OAuth 2.0, IAM Policies, JWT Authentication, Secure RESTful API Design

Automation & Scripting: Python Scripting for AWS, Terraform for Cloud Resource Management, Network Automation using Ansible, pyATS (basic), Netmiko (exposure)

Cloud Monitoring & Troubleshooting: AWS CloudWatch, Prometheus, Grafana

Networking & Security: OAuth 2.0, IAM, SNMP, SSH, HTTPS, NETCONF, API Security

Web Servers & Frameworks: Gunicorn, Uvicorn, WSGI, ASGI

Testing & Debugging: pytest, Unit Testing, Integration Testing, Postman, Automated Testing Strategies

EDUCATION:

- Bachelor's in computer science engineering, Osmania University, India (CGPA-4.0)

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PROFESSIONAL EXPERIENCE:

JPMC, NYC, NY
Sr. Python Developer

June 2023 – Present

Responsibilities:

- Designed and deployed backend microservices using Flask and FastAPI, delivering stateless middle-tier orchestration logic without database dependencies for optimized query execution. Implemented OAuth 2.0, JWT authentication, IAM roles, and API security best practices, reducing unauthorized access risks by 40%. Implemented OAuth 2.0, JWT authentication, and IAM roles to enhance security. Configured AWS VPCs, security groups, and SSL/TLS encryption for API protection.
- Integrated SNMP and SSH-based monitoring, reducing system downtime by 30%. Developed Python scripts for network configuration verification and device state validation as part of infrastructure automation. Collaborated with DevOps to integrate network test cases using Ansible playbooks and basic pyATS test cases.
- Developed and optimized SQL queries for relational databases (PostgreSQL, MySQL), reducing query execution time by 30%. Designed and implemented Python-based RESTful APIs using Django and Flask, ensuring seamless database interactions and improved application performance.
- Built mobile-first, responsive web applications using HTML5, CSS3, and Bootstrap, ensuring compatibility across 95% of devices and browsers.
- Enhanced front-end deliverables by collaborating with UI teams to integrate Angular components, ensuring a seamless user experience and responsiveness across devices.
- Maintained all existing responsibilities (e.g., secure API design, cloud deployments) but add a bullet point highlighting any cross-functional work with front-end frameworks if applicable.
- Created dynamic, responsive user interfaces with JavaScript (ES6+), React.js, React Hooks, and Redux, resulting in a 40% increase in user engagement.
- Deployed services using WSGI and ASGI-compatible servers (Gunicorn, Uvicorn) on AWS Lambda and ECS for scalable, high-performance APIs by 35% and ensuring scalability to handle large user volumes.
- Applied asynchronous programming using Python's AsyncIO and concurrent futures to manage concurrent I/O operations and reduce latency by 20%.
- Designed and deployed distributed computing applications using PySpark on AWS EMR, optimizing data processing performance by 40%, improving Redshift query performance, and overall cutting business report generation time by 30%.
- Implemented and managed Quartz Scheduler jobs as part of a remote risk framework, streamlining batch processing and reducing data latency in trading pipelines by 25%.
- Led a team of 5 developers, managed project timelines, conducted code reviews, and mentored junior engineers. Collaborated with Treasury Tech stakeholders to align trading system requirements with scalable back-end services. Tracked issues and coordinated tasks through JIRA, improving team productivity and communication by 15%.
- Developed enterprise-scale Python applications with SQL-based ETL pipelines, improving data transformation and query execution efficiency.
- Deployed applications on AWS (S3, Lambda, Glue, Redshift, EMR, API Gateway, CloudFormation), ensuring scalability, security, and cost efficiency.
- Designed and deployed RESTful APIs with GraphQL, significantly reducing data retrieval times by 25% and providing enhanced flexibility for front-end teams.
- Architected and implemented containerized microservices on OpenShift and Kubernetes, ensuring high availability, security, and efficient resource management. Designed CI/CD pipelines with Jenkins, GitHub Actions, and SonarQube, improving deployment efficiency by 35%.
- Conducted thorough API testing using Postman, ensuring 100% functionality of RESTful endpoints and reducing post-release issues by 15%.

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- Developed a real-time communication system using WebSocket's, enhancing user engagement by 50% with instant updates and dynamic interactions.
- Optimized data storage and access in MySQL and DynamoDB, improving query performance by 15% through advanced indexing and optimization strategies.
- Streamlined CI/CD pipeline processes with Jenkins and Git, reducing deployment time by 30% and enabling faster, more reliable code releases.
- Strengthened application security by implementing OAuth 2.0 protocols, enhancing authorization processes, and reducing unauthorized access risks by 40%.
- Employed XML, XSLT, and XSD for data transformation and validation, ensuring consistent data flow between systems and improving data integrity by 25%.
- Worked within Agile frameworks (Scrum, Kanban) to ensure 95% on-time delivery of incremental product releases while aligning with stakeholder goals.
- Automated infrastructure deployment and management using CloudFormation and Ansible, cutting deployment times by 40% and minimizing configuration errors.
- Wrote unit tests using PyUnit, achieving 98% test coverage for backend components, and utilized Selenium for end-to-end testing, reducing bugs by 25%.

Environment: Python, Django, Django ORM, HTML5, CSS3, Bootstrap, JavaScript (ES6+), React.js, Node.js, Express.js, NumPy, Pandas, SciPy, JIRA, AWS EC2, S3, RDS, Lambda, Elastic Beanstalk, CloudFront, ECS, EKS, GraphQL, Postman, WebSockets, MySQL, DynamoDB, Jenkins, Git, OAuth 2.0, XML, Agile (Scrum), CloudFormation, Ansible, PyUnit, Selenium.

Xcel Energy, Minneapolis, MN
Python Developer

October 2021 – May 2023

Responsibilities:

- Developed middle-tier microservices in Flask and FastAPI, handling high-throughput request/response workflows without database interaction.
- Implemented asynchronous tasks using Python's AsyncIO and concurrent.futures modules to improve I/O-bound performance across services.
- Deployed backend services on AWS Lambda and ECS using Flask with Gunicorn and Uvicorn servers, improving startup time and concurrency.
- Created unit tests and functional tests using PyTest to ensure service reliability and rapid CI/CD pipeline deployment.
- Led the development of full stack applications by designing robust RESTful APIs in Python and crafting modern front-end interfaces using Angular, TypeScript, and Material-UI to deliver dynamic, user-friendly experiences.
- Developed and optimized SQL queries and indexing strategies for PostgreSQL, Redshift, and PL/SQL, enhancing database performance and reducing query execution time by 30%.
- Implemented GraphQL APIs for real-time data fetching, improving efficiency. Enhanced system monitoring with AWS CloudWatch and logging tools, reducing downtime incidents by 27%.
- Developed and optimized complex SQL queries, stored procedures, and indexing strategies in PostgreSQL and MySQL, reducing query execution time by 30%. Ensured data integrity and reliability with ACID-compliant transactions.
- Developed and deployed microservices-based applications, enhancing system modularity, maintainability, and deployment flexibility, which reduced downtime by 23%.
- Configured and managed AWS API Gateway security policies, ensuring secure API communication, reducing security vulnerabilities by 40%, and implementing Redshift encryption for secure data storage.
- Implemented IAM-based authentication, OAuth 2.0, and JWT token validation to restrict unauthorized access. Deployed AWS security groups and VPCs, enhancing network isolation and reducing data breaches. Used SNMP and SSH for real-time system monitoring, improving issue resolution time by 25%.

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- Created detailed API documentation using Swagger, improving clarity for developers and reducing integration errors by 15%.
- Worked with relational (PostgreSQL) and NoSQL (Redis) databases, enhancing data storage reliability and boosting query performance by 20%.
- Deployed cloud-based solutions with AWS **Glue, Redshift, EMR, EC2, S3, RDS, and Lambda**, optimizing ETL pipelines and reducing infrastructure costs by 25%.
- Tracked tasks and managed projects using JIRA, increasing team productivity by 20% and ensuring timely milestone completion.
- Enhanced SQL query performance for SQL and NoSQL databases, including **Redshift**, reducing query execution times by 20% and optimizing analytical workloads.
- Configured AWS CloudFront to optimize global content delivery, reducing latency by 25% for end users.
- Designed and deployed scalable microservices applications on OpenShift and Kubernetes, improving resource efficiency and security.
- Implemented Infrastructure as Code (IaC) using Terraform, improving deployment efficiency and reducing manual intervention by 35%.
- Monitored system performance with Splunk, optimizing operational efficiency and reducing troubleshooting time by 25%.
- Implemented secure authentication and authorization protocols with OAuth 2.0, enhancing security and decreasing unauthorized access incidents by 30%.
- Implemented CI/CD pipelines with Jenkins, Harness, and Artifactory, ensuring automated deployments with zero downtime. Integrated SonarQube for code quality analysis, reducing security vulnerabilities by 35%.
- Collaborated in Agile sprints and stand-ups, increasing on-time delivery by 15% and improving team cohesion and efficiency.

Environment: Python, FastAPI, SQLAlchemy, Angular, TypeScript, Material-UI, AWS EC2, S3, RDS, Lambda, PostgreSQL, Redis, NoSQL, AWS API Gateway, Swagger, TensorFlow, JIRA, WebSockets, AWS CloudFront, Pandas, NumPy, Docker, Kubernetes, Terraform, Splunk, OAuth 2.0, Jenkins, Agile.

Doodle Blue Innovations, India.
Python Developer

December 2018 – June 2021

Responsibilities:

- Built and maintained Python microservices using Flask, with focus on low-latency, RESTful API endpoints and async HTTP handlers.
- Utilized Python's asyncio and threading libraries to optimize event-driven service responses under concurrent loads.
- Configured Flask applications to run with Gunicorn/Uvicorn for ASGI compatibility and higher throughput.
- Designed stateless services following REST principles and integrated them into orchestration pipelines.
- Developed and deployed robust web applications using Python and Flask, ensuring high-performance back-end systems customized to meet specific user needs.
- Architected and deployed scalable serverless applications using Azure Functions, focusing on cost-effective and efficient solutions for dynamic workloads.
- Integrated RabbitMQ for improved asynchronous message handling, ensuring reliable service communication and enhancing system efficiency.
- Managed identity and access controls with Azure Active Directory (AD), strengthening security and simplifying authentication for cloud applications.
- Designed and implemented efficient database models with SQLAlchemy ORM, facilitating smooth data handling and integration within the system.
- Created and integrated RESTful APIs for seamless communication between front-end and back-end components, enhancing overall system efficiency.

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- Utilized Git for version control, collaborating with cross-functional teams to ensure high-quality code and maintain project timelines and efficiency.
- Developed responsive and user-centric front-end interfaces with Vue.js, HTML5, CSS3, and JavaScript, optimizing user experience across multiple devices.
- Performed unit and integration testing with Mockito, identifying issues and ensuring high-quality standards by resolving defects before production deployment.
- Employed Eclipse IDE for seamless development, debugging, and testing, optimizing the workflow and improving code quality in the development lifecycle.
- Configured Nginx for effective server request management, enhancing application performance, reliability, and availability across the system.
- Applied Agile Kanban principles to manage workflows and meet business goals, delivering projects on time and ensuring high levels of stakeholder satisfaction.
- Built and maintained CI/CD pipelines using Azure DevOps and Maven, automating build, test, and deployment workflows for efficient software delivery.
- Created detailed project documentation and coordinated with teams using Confluence, ensuring transparency and effective communication across all project stages.
- Leveraged Azure Data Factory (ADF), Azure Logic Apps, and Azure App Service to streamline data workflows and enhance cloud-based application deployment processes.

Environment: Python, Flask, Azure Functions, RabbitMQ, Azure Active Directory (AD), SQLAlchemy, RESTful APIs, Git, Vue.js, HTML5, CSS3, JavaScript, Mockito, Eclipse IDE, Nginx, Agile Kanban, Azure DevOps, Maven, Confluence.

Fission Labs, India.
Software Developer

January 2016 – November 2018

Responsibilities:

- Developed backend components for a distributed web platform using Django and Python multithreading to manage high-volume user interactions.
- Applied asynchronous programming concepts to improve backend response times and reduce blocking I/O operations.
- Participated in designing stateless services with minimal reliance on persistent storage layers.
- Built robust web applications using Python and Django, ensuring scalable and maintainable backend systems to support complex user interactions and data processing.
- Designed and optimized MySQL databases, creating efficient data models and writing complex queries to improve data retrieval and storage performance.
- Improved user interaction and front-end performance by utilizing jQuery and AJAX, enhancing the responsiveness and interactivity of the application.
- Performed unit and integration testing with PyTest to identify and resolve bugs, ensuring the delivery of high-quality, production-ready software.
- Integrated secure web services utilizing SOAP and WSDL protocols, ensuring seamless communication between disparate systems while maintaining data integrity.
- Adhered to Agile Scrum methodologies, actively contributing to sprint planning, daily stand-ups, and retrospectives to drive continuous improvements and meet project goals.
- Automated build and deployment processes using ANT and CI/CD tools, streamlining deployments across multiple environments for smoother releases and minimal downtime.
- Implemented version control with Git, maintaining organized repositories and managing collaborative workflows to ensure code integrity and version consistency.
- Developed dynamic, responsive front-end designs with HTML, CSS, and JavaScript, optimizing user experiences for both desktop and mobile platforms.
- Utilized JIRA to track tasks and issues, ensuring efficient project management and prompt issue resolution to meet project deadlines.

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- Managed structured data using XML, enabling cross-platform data sharing and integration to ensure consistency across different systems.

Environment: Python, Django, MySQL, jQuery, AJAX, PyTest, SOAP, WSDL, Git, ANT, CI/CD, HTML, CSS, JavaScript, JIRA, XML.